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System and method of detecting an airflow fault condition

Abstract

A method of detecting an airflow fault condition in a gas turbine engine, the method including: operating the gas turbine engine with a thermal transport bus having an intermediary heat exchange fluid flowing therethrough; determining a performance characteristic of the intermediary heat exchange fluid in the thermal transport bus is outside of a predetermined range, wherein the performance characteristic includes a temperature, a pressure, a flowrate, or a combination thereof; and indicating an airflow fault condition in response to determining the performance characteristic is outside of the predetermined range.

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Background/Summary

FIELD

(1) In general, the present disclosure relates to management of thermal energy in an engine. In particular, the present disclosure relates to a system and method of fault detection using a thermal transport bus in a gas turbine engine.

BACKGROUND

(2) A gas turbine engine generally includes a turbomachine and a rotor assembly. Gas turbine engines, such as turbofan engines, may be used for aircraft propulsion. In the case of a turbofan engine, the rotor assembly may be configured as a fan assembly.

(3) Various accessory systems are included to ensure that the fan and/or core operate as desired. For example, a main lubrication system provides lubrication to, e.g., bearings and gear meshes within a compressor section, a turbine section, and a power gear box (if provided). In addition to the lubricating properties provided to such components, the main lubrication system can be used to remove heat from such components such that they operate within a desired temperature range.

(4) Other accessory systems of the gas turbine engine, such as an environmental control system, also require heat removal during operation. Accordingly, gas turbine engines typically include numerous heat exchangers, each heat exchanger dedicated to an individual accessory system of the gas turbine engine.

(5) In existing thermal energy management systems, thermal transport buses are used to manage the transfer of thermal energy among various heat exchangers. During operation of the gas turbine engine, airflow fault conditions, such as broken pipes, may occur. The inventors of the present disclosure have found that such broken pipes may create various operational issues within the gas turbine engine and thermal energy management systems, and thus improvements to address these issues would be welcomed in the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:
- (2) FIG. 1 is a schematic, cross-sectional view of an exemplary gas turbine engine in accordance with an exemplary aspect of the present disclosure.
- (3) FIG. 2 is a schematic view of a thermal management system in accordance with an exemplary aspect of the present disclosure.
- (4) FIG. 3 is a flowchart of a method of detecting an airflow fault condition of a gas turbine engine in accordance with an exemplary aspect of the present disclosure.
- (5) FIG. 4 is a simplified schematic view of a heat exchanger assembly positioned between a compressor section and a turbine section in accordance with an exemplary aspect of the present disclosure.
- (6) FIG. 5 is a schematic view of another thermal management system in accordance with an exemplary aspect of the present disclosure.
- (7) FIG. 6 is a flowchart of a method of regulating pressure in a thermal transport bus of a gas turbine engine in accordance with an exemplary aspect of the present disclosure.

DETAILED DESCRIPTION

- (8) Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.
- (9) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.
- (10) For purposes of the description hereinafter, the terms “upper”, “lower”, “right”, “left”, “vertical”, “horizontal”, “top”, “bottom”, “lateral”, “longitudinal”, and derivatives thereof shall relate to the illustrated embodiments as oriented in the drawing figures. However, it is to be understood that the embodiments may assume various alternative variations, except where expressly specified to the contrary. It is also to be understood that the specific devices illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the disclosure. Hence, specific dimensions and other physical characteristics related to the embodiments disclosed herein are not to be considered as limiting.
- (11) As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.
- (12) The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.
- (13) The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.
- (14) The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

(15) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

(16) Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 1, 2, 4, 10, 15, or 20 percent margin. These approximating margins may apply to a single value, either or both endpoints defining numerical ranges, and/or the margin for ranges between endpoints.

(17) Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

(18) A “third stream” as used herein means a non-primary air stream capable of increasing fluid energy to produce a minority of total propulsion system thrust. A pressure ratio of the third stream may be higher than that of the primary propulsion stream (e.g., a bypass or propeller driven propulsion stream). The thrust may be produced through a dedicated nozzle or through mixing of an airflow through the third stream with a primary propulsion stream or a core air stream, e.g., into a common nozzle.

(19) In certain exemplary embodiments an operating temperature of the airflow through the third stream may be less than a maximum compressor discharge temperature for the engine, and more specifically may be less than 350 degrees Fahrenheit (such as less than 300 degrees Fahrenheit, such as less than 250 degrees Fahrenheit, such as less than 200 degrees Fahrenheit, and at least as great as an ambient temperature). In certain exemplary embodiments these operating temperatures may facilitate heat transfer to or from the airflow through the third stream and a separate fluid stream. Further, in certain exemplary embodiments, the airflow through the third stream may contribute less than 50% of the total engine thrust (and at least, e.g., 2% of the total engine thrust) at a takeoff condition, or more particularly while operating at a rated takeoff power at sea level, static flight speed, 86 degree Fahrenheit ambient temperature operating conditions.

(20) Furthermore in certain exemplary embodiments, aspects of the airflow through the third stream (e.g., airstream, mixing, or exhaust properties), and thereby the aforementioned exemplary percent contribution to total thrust, may passively adjust during engine operation or be modified purposefully through use of engine control features (such as fuel flow, electric machine power, variable stators, variable inlet guide vanes, valves, variable exhaust geometry, or fluidic features) to adjust or optimize overall system performance across a broad range of potential operating conditions.

(21) The term “turbomachine” or “turbomachinery” refers to a machine including one or more compressors, a heat generating section (e.g., a combustion section), and one or more turbines that together generate a torque output.

(22) The term “gas turbine engine” refers to an engine having a turbomachine as all or a portion of its power source. Example gas turbine engines include turbofan engines, turboprop engines, turbojet engines, turboshaft engines, etc., as well as hybrid-electric versions of one or more of these engines.

(23) The term “combustion section” refers to any heat addition system for a turbomachine. For example, the term combustion section may refer to a section including one or more of a deflagrative combustion assembly, a rotating detonation combustion assembly, a pulse detonation combustion assembly, or other appropriate heat addition assembly. In certain example

embodiments, the combustion section may include an annular combustor, a can combustor, a cannular combustor, a trapped vortex combustor (TVC), or other appropriate combustion system, or combinations thereof.

(24) The terms “low” and “high”, or their respective comparative degrees (e.g., -er, where applicable), when used with a compressor, a turbine, a shaft, or spool components, etc. each refer to relative speeds within an engine unless otherwise specified. For example, a “low turbine” or “low speed turbine” defines a component configured to operate at a rotational speed, such as a maximum allowable rotational speed, lower than a “high turbine” or “high speed turbine” at the engine.

(25) As used herein, the terms “integral”, “unitary”, or “monolithic” as used to describe a structure refers to the structure being formed integrally of a continuous material or group of materials with no seams, connections joints, or the like. The integral, unitary structures described herein may be formed through additive manufacturing to have the described structure, or alternatively through a casting process, etc.

(26) Aspects of the present disclosure present a method of detecting an airflow fault condition in a gas turbine engine and reconfiguring a cooling airflow scheme in response to the airflow fault condition. For example, a thermal transport bus is used to detect an air side broken pipe by sensing a change (e.g., gradual, instantaneous) in a thermal performance characteristic (e.g., temperature, pressure) of an intermediary heat exchange fluid of the thermal transport bus. Also disclosed is a means to divert flow away from a damaged heat exchanger to an operational heat exchanger by way of modifying a configuration of airflow in the gas turbine engine in response to the indicated airflow fault condition, and optionally modifying a configuration of thermal fluid flow within the thermal transport bus. Benefits of the disclosed method include improved fuel burn by the engine, improved reliability of the thermal managements system(s), reduced need for bleed air, and a reduced weight of the engine.

(27) Referring now to the drawings, wherein identical numerals indicate the same elements throughout the figures, FIG. 1 is a schematic, cross-sectional view of a propulsion system 10 in accordance with an exemplary embodiment of the present disclosure. More particularly, for the embodiment of FIG. 1, propulsion system 10 includes a gas turbine engine, referred to herein as “turbofan engine 12.” In one example, turbofan engine 12 can be a high-bypass turbofan jet engine. As shown in FIG. 1, turbofan engine 12 defines an axial direction A (extending parallel to a longitudinal centerline 14 provided for reference) and a radial direction R. In general, turbofan engine 12 includes a fan section 16 and a turbomachine 18 disposed downstream from fan section 16.

(28) The exemplary turbomachine 18 depicted generally includes a substantially tubular outer casing 20 that defines an annular inlet 22. Outer casing 20 encases, in serial flow order/relationship, a compressor section including a booster or low pressure compressor 24 (“LP compressor 24”) and a high pressure compressor 26 (“HP compressor 26”); a combustion section 28; a turbine section including a high pressure turbine 30 (“HP turbine 30”) and a low pressure turbine 32 (“LP turbine 32”). A high pressure shaft or spool 34 (“HP spool 34”) drivingly connects HP turbine 30 to HP compressor 26. A low pressure shaft or spool 36 (“LP spool 36”) drivingly connects LP turbine 32 to LP compressor 24.

(29) For the embodiment depicted, fan section 16 includes a variable pitch fan 38 having a plurality of fan blades 40 coupled to a disk 42 in a spaced apart manner. As depicted, fan blades 40 extend outwardly from disk 42 generally along radial direction R. Each fan blade 40 is rotatable relative to disk 42 about a pitch axis P by virtue of fan blades 40 being operatively coupled to a suitable actuation member 44 configured to collectively vary the pitch of fan blades 40, e.g., in unison. Fan blades 40, disk 42, and actuation member 44 are together rotatable about longitudinal centerline 14 by LP spool 36 across a power gear box 46. Power gear box 46 includes a plurality of gears for stepping down the rotational speed of LP spool 36 to a more efficient rotational fan speed.

(30) Referring still to the exemplary embodiment of FIG. 1, disk 42 is covered by a rotatable front

hub **48** aerodynamically contoured to promote an airflow through the plurality of fan blades **40**. Additionally, fan section **16** includes an annular fan casing or outer nacelle **50** that circumferentially surrounds variable pitch fan **38** and/or at least a portion of turbomachine **18**. It should be appreciated that in some embodiments, nacelle **50** is configured to be supported relative to turbomachine **18** by a plurality of circumferentially spaced outlet guide vanes **52**. Moreover, a downstream section **54** of nacelle **50** extends over an outer portion of turbomachine **18** so as to define a bypass airflow passage **56** therebetween.

(31) During operation of turbofan engine **12**, a volume of air **58** enters turbofan engine **12** through an associated inlet **60** of nacelle **50** and/or fan section **16**. As the volume of air **58** passes across fan blades **40**, a first portion of air **58** as indicated by arrows **62** is directed or routed into bypass airflow passage **56** and a second portion of air **58** as indicated by arrow **64** is directed or routed into LP compressor **24**. The ratio between first portion of air **62** and second portion of air **64** is commonly known as a bypass ratio. The pressure of second portion of air **64** is then increased as second portion of air **64** is routed through high pressure (HP) compressor **26** and into combustion section **28**, where second portion of air **64** is mixed with fuel and burned to provide combustion gases **66**. Subsequently, combustion gases **66** are routed through HP turbine **30** and LP turbine **32**, where a portion of thermal and/or kinetic energy from combustion gases **66** is extracted.

(32) Simultaneously, the pressure of first portion of air **62** is substantially increased as first portion of air **62** is routed through bypass airflow passage **56** before first portion of air **62** is exhausted from a fan nozzle exhaust section **68** of turbofan engine **12**, also providing propulsive thrust.

(33) Moreover, as is depicted schematically, turbofan engine **12** further includes various accessory systems to aid in the operation of turbofan engine **12** and/or an aircraft including turbofan engine **12**. For example, turbofan engine **12** further includes a main lubrication system **70** configured to provide a lubricant to, e.g., various bearings and gear meshes in the compressor section (including LP compressor **24** and HP compressor **26**), the turbine section (including HP turbine **30** and LP turbine **32**), HP spool **34**, LP spool **36**, and power gear box **46**. The lubricant provided by main lubrication system **70** increases the useful life of such components and removes a certain amount of heat from such components.

(34) Additionally, turbofan engine **12** includes a compressor cooling air (“CCA”) system **72** for providing air from one or both of HP compressor **26** or LP compressor **24** to one or both of HP turbine **30** or LP turbine **32**. The CCA system **72** may include a duct and a CCA heat exchanger. The duct may receive an airflow from the compressor section and provide such airflow to the CCA heat exchanger to be cooled. The cooled airflow may then be provided to, e.g., the turbine section to cool various components of the turbine section. Moreover, turbofan engine **12** includes an active thermal clearance control (“ACC”) system **74** for cooling a casing of the turbine section to maintain a clearance between the various turbine rotor blades and the turbine casing within a desired range throughout various engine operating conditions. Although not depicted, the ACC system **74** may similarly include a duct for receiving an airflow and providing such airflow to an ACC heat exchanger. Furthermore, turbofan engine **12** includes a generator lubrication system **76** for providing lubrication to an electronic generator. In one example, the electronic generator provides electrical power to a startup electric motor for turbofan engine **12**, various other electronic components of turbofan engine **12**, and/or an aircraft including turbofan engine **12**.

(35) As is also depicted schematically, turbofan engine **12** drives or enables various other accessory systems for an aircraft including turbofan engine **12**. For example, turbofan engine **12** provides compressed air from the compressor section to an environmental control system (“ECS”) **78**. In one example, ECS **78** provides an air supply to a cabin of the aircraft for pressurization and thermal control. Air can be provided from turbofan engine **12** to an electronics cooling system **80** for maintaining a temperature of certain electronic components of turbofan engine **12** and/or the aircraft within a desired range. Additionally, a surface cooler **82** can be included in fan section **16**. In this example, surface cooler **82** is a heat sink exchanger. Surface cooler **82** is disposed along a

surface (e.g., annular fan casing or outer nacelle **50** that circumferentially surrounds fan **38**) of fan section **16**. In this example, surface cooler **82** is exposed to (e.g., in thermal communication with) first portion of air **62** flowing through turbofan engine **12**. Surface cooler **82** functions by transferring thermal energy from surface cooler **82** into first portion of air **62** either directly via conduction (and/or convection) or indirectly via a sidewall of fan section **16**.

(36) It should be appreciated, however, that turbofan engine **12** depicted in FIG. **1** is by way of example only, and that in other exemplary embodiments, aspects of the present disclosure may additionally, or alternatively, be applied to any other suitable gas turbine engine. For example, in other exemplary embodiments, turbofan engine **12** may instead be any other suitable aeronautical gas turbine engine, such as a turbojet engine, turboshaft engine, turboprop engine, etc. Additionally, in still other exemplary embodiments, turbofan engine **12** may include any other suitable number and/or configuration of shafts, spools, compressors, turbines, etc.; may be configured as a direct drive engine (e.g., excluding power gear box **46**); may be a fixed-pitch fan; may be an unducted turbofan engine (excluding nacelle **50**); etc.

(37) Referring now to FIG. **2**, a schematic, flow diagram is provided of a thermal management system **100** in accordance with an exemplary embodiment of the present disclosure for incorporation at least partially into the exemplary turbofan engine **12** of FIG. **1**.

(38) As shown, thermal management system **100** generally includes a thermal transport bus **102**. Thermal transport bus **102** includes an intermediary heat exchange fluid flowing therethrough and may be formed of one or more suitable fluid conduits. The heat exchange fluid is disposed to flow through thermal transport bus **102**. The heat exchange fluid may be an incompressible fluid having a high temperature operating range. For example, in certain embodiments, the heat exchange fluid may be a liquid such as a water and ethanol mixture, or any suitable dielectric fluid. In other embodiments, however, the heat exchange fluid may be any other suitable fluid, such as an oil having a relatively high temperature operating range, a phase change fluid (configured to change between, e.g., liquid and gas phases across the anticipated operating temperature ranges), a supercritical heat exchange fluid (such as a supercritical CO₂), etc.

(39) Thermal transport bus **102** also includes a first flow loop **104**. First flow loop **104** is a closed loop conduit containing the intermediary heat exchange fluid flowing therethrough. First flow loop **104** includes at least one of first heat exchangers **106a-e**. First heat exchangers **106a-e** are devices for transferring thermal energy between two fluids.

(40) In certain exemplary embodiments, first heat exchangers **106a-b** may be heat source heat exchangers. More specifically, in at least certain exemplary aspects, first heat exchanger **106a** may be a compressor bleed air heat exchanger and heat exchanger **106b** may be a compressor discharge pressure heat exchanger. Additionally, or alternatively, in other exemplary embodiments, first heat exchangers **106c-e** may be heat sink heat exchangers. More specifically, in at least certain exemplary aspects, first heat exchanger **106c** may be a flowpath stream heat sink heat exchanger, first heat exchanger **106d** may be a fuel cooled bus cooler heat sink heat exchanger, and first heat exchanger **106e** may be an air cooled bus cooler.

(41) It will be appreciated, however, that in other exemplary embodiments, any of first heat exchangers **106a-e** may be configured as a main lubrication system heat exchanger, a cooled cooling air system heat exchanger, an active thermal clearance control system heat exchanger, a generator lubrication system heat exchanger, an environmental control system heat exchanger, an electronics cooling system heat exchanger, an air cooled bus cooler system heat exchanger, a fuel cooled bus cooler system heat exchanger, a compressor discharge pressure system heat exchanger, or a waste heat recovery system heat exchanger.

(42) First flow loop **104** additionally includes upstream valves **108** and downstream valves **110**. In certain exemplary embodiments, upstream valves **108** and downstream valves **110** may be variable two-way valves with an inlet and an outlet fluidly connected with thermal transport bus **102**.

Additionally, or alternatively, in other exemplary embodiments, upstream valves **108** and

downstream valves **110** may be three-way heat sink valves with an inlet fluidly connected with thermal transport bus **102**, a first outlet fluidly connected with thermal transport bus **102**, and a second outlet fluidly connected with a bypass line configured to transport flow of the heat exchange fluid around a given heat exchanger and back into thermal transport bus **102**.

(43) First flow loop **104** further includes sensors **112**. Sensors **112** are devices configured to detect or measure a property of a medium. In certain exemplary embodiments, sensors **112** can be configured to sense, detect, monitor, or any combination thereof a performance characteristic such as a temperature, a pressure, a density, a flow rate, a flow direction, or other physical (or chemical) properties of the heat exchange fluid in first flow loop **104** and in a second flow loop **118** (described below).

(44) First flow loop **104** also includes a first pump **114** to move the heat exchange fluid through first flow loop **104**. First pump **114** is fluidly connected to first flow loop **104** and is disposed to generate a closed-loop flow of the heat exchange fluid in thermal transport bus **102** within first flow loop **104**.

(45) First pump **114** includes a first compressor **116**. First compressor **116** is a device configured to compress or otherwise generate a flow of the heat exchange fluid in thermal transport bus **102**, and in particular in first flow loop **104**. First compressor **116** is fluidly connected to and disposed between heat exchanger **106e** and heat exchanger **106a**.

(46) Thermal transport bus **102** also includes a second flow loop **118**. Similar to first flow loop **104**, second flow loop **118** also is a closed loop conduit containing the intermediary heat exchange fluid flowing therethrough.

(47) Second flow loop **118** includes at least one second heat exchanger **120a-b**. In certain exemplary embodiments, second heat exchangers **120a-b** may be heat source heat exchangers. More specifically, in at least certain exemplary aspects, second heat exchanger **120a** may be a compressor bleed air heat exchanger and second heat exchanger **120b** may be a compressor discharge pressure heat exchanger. Second heat exchanger **120a** is in fluid communication with and is disposed downstream from a second pump **126** (described below). Second heat exchanger **120b** is in fluid communication with and is disposed downstream from second heat exchanger **120a**.

(48) In certain exemplary embodiments, any one or more of heat exchangers **106a-e** and **120a-b** may be configured as a heat sink exchanger for transferring heat from the heat exchange fluid in thermal transport bus **102**, e.g., to atmosphere, to fuel, to a fan stream, etc. For example, in certain embodiments, a heat sink exchanger (e.g., one or more of heat exchangers **106a-e** and **120a-b**) may include at least one of a RAM heat exchanger, a fuel heat exchanger, a fan stream heat exchanger, or a bleed air heat exchanger. The RAM heat exchanger may be configured as an “air to heat exchange fluid” heat exchanger integrated into one or both of turbofan engine **12** or an aircraft including turbofan engine **12**. During operation, the RAM heat exchanger may remove heat from any heat exchange fluid therein by flowing a certain amount of RAM air over the RAM heat exchanger. Additionally, the fuel heat exchanger is a “liquid to heat exchange fluid” heat exchanger wherein heat from the heat exchange fluid is transferred to a stream of liquid fuel for turbofan engine **12**. Moreover, the fan stream heat exchanger is generally an “air to heat exchange fluid” heat exchanger which flows, e.g., bypass air over heat exchange fluid to remove heat from the heat exchange fluid. Further, the bleed air heat exchanger is generally an “air to heat exchange fluid” heat exchanger which flows, e.g., bleed air from LP compressor **24** over the heat exchange fluid to remove heat from the heat exchange fluid. An engine including one or more of these heat exchangers may include one or more ducts to provide the cooling fluid (e.g., air, fuel, etc.) to and from the heat exchangers, or the heating fluid (e.g., air, oil, etc.) to and from the heat exchangers. As used herein, the term “duct” refers generally to any conduit or enclosure designed for, or capable of, providing a flow of fluid between two locations.

(49) Second flow loop **118** also includes upstream valves **122** and downstream valves **124**. Upstream valves **122** and downstream valves **124** are devices for controlling an amount of a fluid

therethrough.

(50) In certain exemplary embodiments, upstream valves **122** and downstream valves **124** may be variable two-way valves with an inlet and an outlet fluidly connected with thermal transport bus **102**. Additionally, or alternatively, in other exemplary embodiments, upstream valves **122** and downstream valves **124** may be three-way heat sink valves with an inlet fluidly connected with thermal transport bus **102**, a first outlet fluidly connected with thermal transport bus **102**, and a second outlet fluidly connected with a bypass line configured to transport flow of the heat exchange fluid around a given heat exchanger and back into thermal transport bus **102**.

(51) Second flow loop **118** additionally includes second pump **126** to move the heat exchange fluid through second flow loop **118**. Second pump **126** is fluidly connected to second flow loop **118** and is disposed to generate a closed-loop flow of the heat exchange fluid in thermal transport bus **102** within second flow loop **118**. In certain exemplary embodiments, first pump **114** and second pump **126** may each be a rotary pump including an impeller, or alternatively may be any other suitable fluid pump.

(52) Second pump **126** includes a second compressor **128**. Second compressor **128** is a device configured to compress or otherwise generate a flow of the heat exchange fluid in thermal transport bus **102**, and in particular in second flow loop **118**. Second compressor **128** is fluidly connected to and disposed between heat exchanger **106e** and heat exchanger **120a**.

(53) In certain exemplary embodiments, first pump **114** and second pump **126** may be powered by an electric motor, or alternatively may be in mechanical communication with and powered by, e.g., the HP shaft **34** or the LP shaft **36** of turbofan engine **12**. More specifically, in at least certain exemplary aspects, thermal management system **100** may include a common drive system **130** to drive first pump **114** and second pump **126**. Common drive system **130** can include an inverter **132** for converting electricity from common drive system **130** from DC electric current to AC electric current.

(54) In certain exemplary embodiments, such as depicted in FIG. 2, It will be appreciated that first flow loop **104** and second flow loop **118** of thermal management system **100** provide redundant and isolated heat removal capacity to a plurality of common heat sources **136A-B**. More specifically, in at least certain exemplary aspects, heat exchanger **106a** and heat exchanger **120a** are configured with a common heat source **136A** such that each of heat exchanger **106a** and heat exchanger **120a** provide independent and isolated heat removal capability (via the heat exchange fluid) to the same common heat source **136A**. Likewise, heat exchanger **106b** and heat exchanger **120b** are configured with a common heat source **136B** such that each of heat exchanger **106b** and heat exchanger **120b** provide independent and isolated heat removal capability (via the heat exchange fluid) to the same common heat source **136B**.

(55) In certain exemplary embodiments, the common heat source **136A** and the common heat source **136B** may be: a main lubrication system heat exchanger for transferring heat from the main lubrication system **70**; a cooled cooling air system heat exchanger for transferring heat from the CCA system **72**; an active thermal clearance control system heat exchanger for transferring heat from the ACC system **74**; a generator lubrication system heat exchanger for transferring heat from the electric machine thermal system **76**; an environmental control system heat exchanger for transferring heat from the ECS **78**; an electronics cooling system heat exchanger for transferring heat from the electronics cooling system **80**; an air cooled bus cooler system heat exchanger; a fuel cooled bus cooler system heat exchanger; a compressor discharge pressure system heat exchanger; or a waste heat recovery system heat exchanger.

(56) First flow loop **104** is isolated from second flow loop **118** such that the heat exchange fluid moving through the first flow loop **104** does not mix with the heat exchange fluid moving through second flow loop **118**. For example, although heat exchanger **106a** and heat exchanger **120a** are configured to remove heat from the same common heat source **136A**, heat exchangers **106a** and **120a** are structurally independent and fluidly isolated such that the heat exchange fluid moving

through the respective heat exchangers **106a** and **120a** does not mix. Likewise, heat exchangers **106b** and **120b** are structurally independent and fluidly isolated such that the heat exchange fluid moving through the respective heat exchangers **106b** and **120b** does not mix.

(57) In certain exemplary embodiments, heat exchangers **106c-e** are in fluid communication with first flow loop **104** and with second flow loop **118**. Additionally, or alternatively, in such exemplary embodiments, first flow loop **104** can be out of fluid communication with heat exchangers **120a** and **120b**, while second flow loop **118** can be out of fluid communication with heat exchangers **106a** and **106b**. It will be appreciated, however, that in other exemplary embodiments, that any of heat exchangers **106a-e** and **120a-b** can be in or out of fluid communication with first flow loop **104** and/or second flow loop **118**.

(58) Thermal management system **100** additionally includes common cold destinations **138A-b**. Common cold destinations **138A-b** are locations or portions of propulsion system **10** (FIG. 1) that make use of a cooling air flow to reduce an amount of thermal energy therein. In certain exemplary embodiments, common cold destination **138A** may be a cooled cooling air source and common cold destination **138B** may be a portion of HP turbine **30** (e.g., a first stage blade of HP turbine **30**).

(59) As is depicted, the engine including the thermal management system **100** may include one or more ducts to transport the heating fluid to the heat exchangers **106a**, **106b**, **120a**, **120b** from the heat sources **136A**, **136B** and to the cold destinations **138A**, **138B** from the heat exchangers **106a**, **106b**, **120a**, **120b**.

(60) Thermal transport bus **102** additionally includes a sensor **140**. In this exemplary embodiment, sensor **140** is connected to and is disposed to monitor a temperature, a pressure, a density, a flow rate, a flow direction, another physical (or chemical) property, or any combination thereof of common cold destination **138B** or of a fluid flowing therethrough.

(61) In other exemplary embodiments, thermal management system **100** may include one or more sensors **140** disposed to sense, detect, or monitor a temperature, a pressure, a density, a flow rate, a flow direction, another physical (or chemical) property, or any combination thereof of common cold destination **138A**, a heat source **142**, a cold source **144**, a cold source **146**, or of a fluid flowing therethrough.

(62) In certain exemplary embodiments, sensors **112** and sensor **140** are configured to transmit a wireless signal to a location in an engine or an aircraft, such as to the cockpit or to a controller **164**. As described below, the controller **164** may be an engine controller (such as a Full Authority Digital Engine Control controller) or an aircraft controller.

(63) Heat source **142** is a fluid source with an amount of thermal energy greater than the amount of thermal energy of the heat exchanger fluid flowing through heat exchanger **106c**. In this exemplary embodiment, heat source **142** is fluidly connected to heat exchanger **106c**. In certain exemplary embodiments, heat source **142** can be a source of exhaust of the engine.

(64) Cold source **144** and cold source **146** are fluid sources with an amount of thermal energy less than the amount of thermal energy of the heat exchanger fluid flowing through heat exchanger **106d** and heat exchanger **106e**, respectively. In certain exemplary embodiments, cold source **144** may be a fuel source of propulsion system **10** (FIG. 1) and cold source **146** may be a working air stream (e.g., third stream flow of air) of propulsion system **10**.

(65) Thermal transport bus **102** additionally includes bypass lines **154A-C**. In certain exemplary embodiments, bypass lines **154A-C** are pipes or conduits configured to transport or divert a flow of the heat exchanger fluid around a component (e.g., any of heat exchangers **106a-e** and **120a-b**) along thermal transport bus **102**.

(66) Bypass line **154A** is fluidly connected to second flow loop **118** at one of upstream valves **122** and at one of downstream valves **124**. More specifically, in at least certain exemplary aspects, bypass line **154A** is configured to divert a flow of the heat exchange working fluid from second flow loop **118**, through bypass line **154A**, around heat exchanger **120a**, and rejoin the flow of the heat exchange fluid back into second flow loop **118**.

(67) Bypass line **154B** is fluidly connected to first flow loop **104** and to second flow loop **118** at upstream valves **122** (directly upstream from heat exchanger **106c**) and at downstream valves **124** (directly downstream from heat exchanger **106c**). More specifically, in at least certain exemplary aspects, bypass line **154B** is configured to divert a flow of the heat exchange working fluid from either first flow loop **104** or second flow loop **118**, through bypass line **154B**, around heat exchanger **106c**, and rejoin the flow of the heat exchange fluid back into first flow loop **104** or second flow loop **118**.

(68) Bypass line **154C** is fluidly connected to first flow loop **104** and to second flow loop **118** at upstream valves **122** (directly upstream from heat exchanger **106e**) and at downstream valves **124** (directly downstream from heat exchanger **106e**). More specifically, in at least certain exemplary aspects, bypass line **154C** is configured to divert a flow of the heat exchange working fluid from either first flow loop **104** or second flow loop **118**, through bypass line **154C**, around heat exchanger **106e**, and rejoin the flow of the heat exchange fluid back into first flow loop **104** or second flow loop **118**.

(69) As depicted in FIG. 2, it may be desired to configure each of heat exchangers **106a-e** and **120a-b** with a respective bypass line **154**, upstream valve **108**, and downstream valve **110**. This provides thermal management system **100** with the capability to isolate one of the heat exchangers configured with each of common heat sources **136A-b** in the event of a failure (e.g., leak or rupture). For example, if heat exchanger **106a** fails or if a component connected to heat exchanger **106a** fails, heat exchanger **106a** can be isolated and bypassed within first flow loop **104** leaving heat exchanger **120a** to provide heat removal capability to the common heat source **136A**. Likewise, if heat exchanger **106c** fails, heat exchanger **106c** can be isolated and bypassed within second flow loop **118**.

(70) Still referring to the embodiment of FIG. 2, the plurality of heat exchangers **106a-e** in first flow loop **104**, and the plurality of heat exchangers **120a-b** in second flow loop **118**, may be configured for selective activation of any combination thereof within their respective flow loop. For this, each of heat exchangers **106a-e** and **120a-b** may be provided with a bypass line **154A-C**, upstream valve **108**, **122**, and downstream valve **110**, **124**. In certain exemplary embodiments, upstream valves **108** may be a three-way heat sink valve with an inlet fluidly connected with thermal transport bus **102**, a first outlet fluidly connected with thermal transport bus **102**, and a second outlet fluidly connected with bypass line **154**. Upstream valve **108** may each be a variable throughput three-way valve, such that upstream valve **108** may vary a throughput from the inlet to the first and/or second outlets. For example, upstream valve **108** may be configured for providing anywhere between zero percent (0%) and one hundred percent (100%) of the heat exchange fluid from the inlet to the first outlet, and similarly, upstream valve **108** may be configured for providing anywhere between zero percent (0%) and one hundred percent (100%) of the heat exchange fluid from the inlet to the second outlet.

(71) In certain exemplary embodiments, the various valves, such as upstream valves **108**, downstream valves **110**, upstream valves **122**, and downstream valves **124**, may be selectively controlled to vary the throughput through the respective valves **108**, **110**, **122**, and **124** of the heat exchange fluid. More specifically, in at least certain exemplary aspects, thermal management system **100** includes controller **164**, such as an engine controller for propulsion system **10** (FIG. 1) (e.g., a Full Authority Digital Engine Control (FADEC) controller), an aircraft controller, a controller dedicated to thermal management system **100**, etc., to facilitate modulation of the respective valves **108**, **110**, **122**, and **124**.

(72) Thermal management system **100** further includes controller **164**. In certain exemplary embodiments, controller **164** may be configured to receive data indicative of various operating conditions and parameters of thermal management system **100** (and of propulsion system **10**) during operation of turbofan engine **12**. For example, in addition to sensors **112** and sensor **140** of thermal management system **100**, propulsion system **10** (FIG. 1) may include one or more sensors

configured to sense data indicative of various operating conditions and parameters of turbofan engine **12** (FIG. **1**), such as throttle setting, temperatures, pressures, etc.

(73) Referring particularly to the operation of controller **164**, in at least certain embodiments, controller **164** can include one or more computing device(s) **166**. The computing device(s) **166** can include one or more processor(s) **166A** and one or more memory device(s) **166B**. The one or more processor(s) **166A** can include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, and/or other suitable processing device. The one or more memory device(s) **166B** can include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, RAM, ROM, hard drives, flash drives, and/or other memory devices.

(74) The one or more memory device(s) **166B** can store information accessible by the one or more processor(s) **166A**, including computer-readable instructions **166C** that can be executed by the one or more processor(s) **166A**. The instructions **166C** can be any set of instructions that when executed by the one or more processor(s) **166A**, cause the one or more processor(s) **166A** to perform operations. In some embodiments, the instructions **166C** can be executed by the one or more processor(s) **166A** to cause the one or more processor(s) **166A** to perform operations, such as any of the operations and functions for which controller **164** and/or the computing device(s) **166** are configured, the operations for modulating a valve as described herein, and/or any other operations or functions of the one or more computing device(s) **166**. The instructions **166C** can be software written in any suitable programming language or can be implemented in hardware. Additionally, and/or alternatively, the instructions **166C** can be executed in logically and/or virtually separate threads on processor(s) **166A**. The memory device(s) **166B** can further store data **166D** that can be accessed by the processor(s) **166A**. For example, the data **166D** can include data indicative of power flows, data indicative of turbofan engine **12** (FIG. **1**) aircraft operating conditions, and/or any other data and/or information described herein.

(75) The computing device(s) **166** can also include a network interface **166E** used to communicate, for example, with the other components of propulsion system **10**, the aircraft incorporating propulsion system **10**, thermal management system **100**, etc. For example, in the embodiment depicted, controller **164** is operably coupled to valves **108**, **110**, **122**, and **124** (as well as, e.g., one or more sensors for sensing data indicative of one or more parameters of propulsion system **10** (FIG. **1**) and/or various accessory systems) through, e.g., the network interface **166E**, such that controller **164** may receive data indicative of various operating parameters sensed by the one or more sensors during operation, various operating conditions of the components, etc., and further may provide commands to control flow of the heat exchange fluid and other operating parameters of these systems, e.g., in response to the data sensed by the one or more sensors and other conditions.

(76) The network interface **166E** can include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, and/or other suitable components. In certain exemplary embodiments, in the embodiment shown, the network interface **166E** is configured as a wireless communication network wirelessly in communication with these components (as is indicated by the dashed communication lines in FIGS. **3A** and **3B**).

(77) The technology discussed herein refers to computer-based systems and actions taken by and information sent to and from computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. In certain exemplary embodiments, processes discussed herein can be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications can be implemented on a single system or distributed across multiple systems. Distributed components can operate sequentially or in parallel.

(78) It will be appreciated that although the exemplary embodiment depicted in FIG. 2 includes upstream valves **108** and downstream valves **110** for bypassing the respective heat exchanger **106a-e** and upstream valves **122** and downstream valves **124** for bypassing the respective heat exchanger **120a-b**, in other embodiments the gas turbine engine and/or thermal management system **100** may have any other suitable configuration for bypassing one or more of these heat exchangers.

(79) In certain exemplary embodiments, the heat source and/or heat sink system may be configured to bypass the heat exchangers of the thermal management system. More specifically, in at least certain exemplary aspects, when fuel is the heat sink thermally coupled to one of heat exchangers **106a-e**, the fuel may be bypassed around the respective heat exchangers **106a-e** with which it is otherwise fluidly coupled. Similarly for example, when a lubrication oil system is a heat source thermally coupled to one of heat exchangers **106a-e**, the lubrication oil may be bypassed around the respective heat exchangers **106a-e** with which it is otherwise fluidly coupled. In such a manner, it may be possible to adjust heat source systems and/or heat sink systems of the aircraft engine in response to a leak or failure or a component, while still allowing for desired operations of thermal management system **100**. For example, it may be possible to shut down a bleed port flow (which may be a heat sink system with respect to thermal management system **100**) and de-rate turbofan engine **12** (FIG. 1) in the event of a malfunctioning heat source pipe. Such may provide for a simpler approach to bypassing heat exchangers when it is determined to be necessary or desirable.

(80) Notably, upstream valve **108** may be in operable communication with controller **164** of turbofan engine **12** (FIG. 1) and/or of an aircraft including turbofan engine **12**. Controller **164** may bypass one or more of heat exchangers **106a-e** in first flow loop **104** and the heat exchangers **120a-b** in second flow loop **118** based on, e.g., a change (e.g., a gradual or sudden change) in an operating condition of turbofan engine **12** and/or aircraft, a temperature or pressure of the heat exchange fluid of thermal management system **100**, and/or any other suitable variables.

(81) In certain exemplary embodiments, a method of detecting an airflow fault condition in turbofan engine **12** (FIG. 1) is provided. More particularly, referring now to FIG. 3, a flow diagram of a method of detecting an airflow fault condition in a gas turbine engine is provided at method **170**. Method **170** may be used with the turbofan engine **12** and the thermal management system described above, or with any other suitable gas turbine engine and thermal management system.

(82) Method **170** includes at **172** operating the gas turbine engine (e.g., the turbofan engine **12** of FIG. 1) with a thermal transport bus (such as thermal transport bus **102**) having an intermediary heat exchange fluid flowing therethrough. Operating the gas turbine engine at **172** includes, for the exemplary aspect depicted, at **174** providing the intermediary heat exchange fluid to a heat exchanger of the thermal transport bus, and at **176** providing a fluid flow through a duct of the gas turbine engine to the heat exchanger to exchange heat with the intermediary heat exchange fluid. For example, providing the intermediary heat exchange fluid to a heat exchanger of the thermal transport bus may include, e.g., providing the heat exchange fluid to one of heat exchangers **106a-e** (see FIG. 2), and providing the fluid flow through the duct of the gas turbine engine to the heat exchanger may include, e.g., providing a fluid flow from one or more of heat sources or cold sources **136A-b**, **142**, **144**, **146**.

(83) The exemplary method **170** depicted further includes at **178** monitoring a performance characteristic of the intermediary heat exchange fluid in thermal transport bus; and at **180** determining the performance characteristic of the intermediary heat exchange fluid in the thermal transport bus is outside of a predetermined range. The performance characteristic can include a temperature, a pressure, a flowrate, or a combination thereof of the intermediary heat exchange fluid. In certain exemplary embodiments, the performance characteristic may be an absolute temperature, pressure, and/or flowrate (e.g., the characteristic at a single location), or may be a differential temperature, pressure, and/or flowrate (e.g., a difference in the characteristic between two locations). The differential measurements may be across, e.g., a heat exchanger, or alternatively may be a differential measurement from a location on or within the thermal transport

bus **102** to another engine location, such as a flowpath characteristic of the engine, such as a turbine inlet temperature or pressure, a compressor exit temperature or pressure, etc.

(84) Monitoring the performance characteristic of the intermediary heat exchange fluid at **178** can include sensing data indicative of the performance characteristic of the intermediary heat exchange fluid with sensor **112** (FIG. 2). For example, in certain exemplary aspects, monitoring the performance characteristic of the intermediary heat exchange fluid at **178** can include measuring the performance characteristic with one or more of sensors **112** (e.g., with a temperature sensor, a pressure sensor, or both of a temperature sensor and a pressure sensor).

(85) Referring still to the exemplary aspect of method **170** depicted in FIG. 3, monitoring the performance characteristic of the intermediary heat exchange fluid at **178** can additionally, or alternatively, include monitoring for an instantaneous change in the performance characteristic. The instantaneous change in the performance characteristic can be defined by when the performance characteristic changes at a rate of 20% or more of a normal operating value of the performance characteristic per 30 seconds. The instantaneous change in the performance characteristic can additionally, or alternatively, be defined by a step function of the performance characteristic. As used herein, the term “step function” refers to a series of intervals each with a substantially constant value that is different than a previous or following interval. A data set can be created that is representative of the monitored performance characteristic of the intermediary heat exchange fluid. In such an embodiment, determining the performance characteristic is outside of the predetermined range at **180** can additionally, or alternatively, include identifying a portion of the data defining a step function. Furthermore, it will be appreciated that with such an exemplary aspect, sensing the performance characteristic of the intermediary heat exchange fluid can include sensing a piecewise change in the performance characteristic of the intermediary heat exchange fluid. As used herein, the term “piecewise change” refers to a function defined by multiple different intervals of with each interval including changing values.

(86) Moreover, monitoring the performance characteristic of the intermediary heat exchange fluid at **178** can additionally, or alternatively, include monitoring for a gradual change in the performance characteristic. The gradual change in the performance characteristic can be defined by when the performance characteristic changes at a rate of 20% or less of a normal operating value of the performance characteristic per 30 seconds.

(87) As noted above, the exemplary method depicted further includes determining the performance characteristic of the intermediary heat exchange fluid in the thermal transport bus is outside of the predetermined range. More specifically, for the exemplary aspect depicted, method **170** includes at **182** determining the monitored performance characteristic of the intermediary heat exchange fluid in the thermal transport bus is outside of the predetermined range.

(88) In response to determining the monitored performance characteristic is outside of the predetermined range at **182**, method **170** includes at **184** indicating an airflow fault condition. In at least certain exemplary aspects, indicating the airflow fault condition at **184** may include providing an audible or visual indicator to a user, providing an electronic signal to a controller, or both.

(89) Moreover, in certain exemplary aspects, the airflow fault condition can be a broken pipe of turbofan engine **12** (FIG. 1). For example, the airflow fault condition may refer a fault external to thermal transport bus **102**, such as a leak, rupture, or the like in the duct of turbofan engine **12** providing the fluid flow to the heat exchanger of thermal management system **100** (FIG. 2) to exchange heat with the intermediary heat exchange fluid of thermal management system **100**. In such a manner, it will be appreciated that when, e.g., a cooling airflow is being provided to a heat sink heat exchanger of thermal transport bus **102** (FIG. 2), if there is a fault in the duct providing the cooling airflow to the heat sink heat exchanger of thermal transport bus **102**, a performance characteristic of the intermediary heat exchange fluid through thermal transport bus **102** will be affected. If the fault is a large leak or rupture, the change in the performance characteristic may be relatively drastic (e.g., an instantaneous change). If the fault is a small leak or rupture, the change

in the performance characteristic may be less drastic (e.g., a gradual change).

(90) Referring still to FIG. 3, method **170** further includes at **186** modifying a configuration of airflow in gas turbine engine in response to the indicated airflow fault condition. In at least certain exemplary aspects, modifying the configuration of airflow in the gas turbine engine at **186** may include modifying the airflow through the duct of the gas turbine engine to the heat exchanger. For example, in certain exemplary aspects, the heat exchanger may be a first heat exchanger, and modifying the configuration of airflow in the gas turbine engine at **186** may include shutting down an airflow through at least a portion of the duct of the gas turbine engine and providing an airflow to a second heat exchanger. The airflow provided may include at least a portion of the airflow previously provided through the duct, and the second heat exchanger may be a redundant heat exchanger to the first heat exchanger.

(91) Additionally, modifying the configuration of airflow in the gas turbine engine can include selectively deactivating fluid communication of heat exchanger (e.g., first heat exchanger) with thermal transport bus, and more specifically can include selectively bypassing the heat exchanger by sending a flow of heat exchanger fluid through the thermal transport bus through a bypass line to the second heat exchanger that is in fluid communication with thermal transport bus.

(92) For example, referring briefly back to the embodiment of FIG. 2, in the embodiment shown, modifying the configuration of airflow in turbofan engine **12** (FIG. 1) can include modifying a configuration of thermal transport bus **102** by varying a position (e.g., opening, closing, partially opening, partially closing, etc.) of one or more of upstream valves **108**, downstream valves **110**, upstream valves **122**, and downstream valves **124** valve that are in fluid communication with thermal transport bus **102**. Modifying the configuration of airflow in turbofan engine **12** can additionally, or alternatively, include modifying a configuration of a secondary circuit of turbofan engine **12** by varying a position of a valve in fluid communication with the secondary circuit. In certain exemplary embodiments, a secondary circuit of turbofan engine **12** can include a circuit not directly connected to either of first flow loop **104** or second flow loop **118**. More specifically, in at least certain exemplary aspects, a secondary circuit may include a compressor bleed air circuit, a cooling air circuit for HP turbine **30**, an amount of air dumped into ambient or into the exhaust stream or combustion gases **66**, etc.

(93) Thermal management system **100** may more efficiently remove heat from the various accessory systems of turbofan engine **12** (FIG. 1) and/or the aircraft during an occurrence of a fault condition such as a leak or a pipe failure. Due to the redundant and selective configuration of the various heat exchangers **106a-e** and **120a-b** for the embodiment depicted, utilization of a certain fuel heat exchanger may be ceased if a leak occurs.

(94) For example, when a fault or pipe failure occurs in a component connected to a specific heat exchanger (e.g., heat exchangers **106a-e** or heat exchangers **120a-b**) of thermal transport bus **102**, that specific heat exchanger can be bypassed in order to utilize, for thermal energy transfer, the remaining heat exchangers that are not in communication with the broken or failed pipe. In this way, the effectiveness of thermal heat transfer of thermal transport bus **102** can be improved in comparison to a failure mode where thermal energy management is less efficient due to a leak caused by the broken or failed pipe.

(95) Benefits of the disclosed method include improved reliability of thermal management system **100** (FIG. 2), a reduced need for bleed air, a reduced weight of propulsion system **10** (FIG. 1), and improved fuel burn by propulsion system **10**.

(96) Referring now to FIG. 4, FIG. 4 is a simplified schematic view of a heat exchanger assembly **200** positioned between a compressor section **202** and a turbine section **204** in accordance with an exemplary aspect of the present disclosure.

(97) In certain exemplary embodiments, compressor section **202** and turbine section **204** may correspond to and be configured in substantially the same manner as the compressor section (e.g., LP compressor **24** and HP compressor **26**) and the turbine section (e.g., HP turbine **30** and LP

turbine 32) discussed with respect to FIG. 1 above.

(98) Heat exchanger assembly **200** includes a first heat exchanger **206** and a second heat exchanger **208**. First heat exchanger **206** and second heat exchanger **208** may be configured in substantially the same manner as described with respect to any of heat exchangers **106a-e** and **120a-b**, respectively, in regard to FIG. 2. Here, first heat exchanger **206** is directly connected to turbine section **204** and to compressor section **202** via a first duct **205** and a first valve **210** positioned in airflow communication with first duct **205**. Likewise, second heat exchanger **208** is directly fluidly connected to turbine section **204** and to compressor section **202** via a second duct **207** and a second valve **212** positioned in airflow communication with second duct **207**. Additionally, first valve **210** is in fluid communication with second valve **212** via a bypass line **216**.

(99) First valve **210** and second valve **212** are capable of occupying a fully open position, a fully closed position, or a position between fully open and fully closed. As used herein with respect to FIG. 3, the term “fully closed” can mean that the normal fluid pathway to first heat exchanger **206** via first duct **205** is closed while the fluid pathway from first valve **210** to bypass line **216** is open. In this way, a fluid passing through heat exchanger assembly **200** can be redirected by one of first valve **210** and/or second valve **212** through bypass line **216** to a different part of heat exchanger assembly **200**.

(100) For example, in this exemplary embodiment, first valve **210** is shown in a fully closed position. With first valve **210** in a fully closed position, the fluid flow path passing through first valve **210** and through first duct **205** becomes closed and the fluid flow through first valve **210** is redirected to bypass line **216**. The redirected flow of fluid from first valve **210** passes through bypass line **216**, through (open) second valve **212**, and onward to second heat exchanger **208** through second duct **207**.

(101) During operation, an open or closed position of first valve **210** (or second valve **212**) can be adjusted in response to a signal from a controller (e.g., controller **164** discussed with respect to FIG. 2). The signal from the controller to open or close first valve **210** can be sent by the controller in response to an indication of an airflow fault condition (such as a failed air pipe), which may in turn be in response to determining a performance characteristic of an intermediary heat exchange fluid through the first heat exchanger **206** is above a predetermined threshold. For example, the signal may be in response to an indication of a step change (e.g., in pressure or temperature) or an unexpected thermal performance of heat exchanger assembly **200** or of another component or system connected to a component of heat exchanger assembly **200**. For example, first heat exchanger **206** and second heat exchanger **208** may be connected to a thermal transport bus **218** of an aircraft (see e.g., thermal transport bus **102** described with respect to FIG. 2). In certain exemplary embodiments, thermal transport bus **218** may be configured in substantially the same manner as thermal transport bus **102** discussed with respect to FIG. 2.

(102) For example, the first heat exchanger **206** may be a heat source heat exchanger, such as a CCA heat exchanger. In such a manner, the embodiment of FIG. 3 may be configured to bleed compressed air from the compressor through first and second ducts **205**, **207**. The compressed air may be provided through first and second ducts **205**, **207** to the first and second heat exchangers **206**, **208**, whereby the compressed air is cooled by respective intermediary heat exchange fluids. The cooled air may then be provided to the turbine section **204** to cool one or more components of the turbine section. In response to determining a performance characteristic of the intermediary heat exchange fluid through the first heat exchanger **206** is above the predetermined threshold, the system may determine there is an airflow fault condition in the first duct, and in response redirect an airflow through the first duct to the second duct using valves **210**, **212**. Optionally, the intermediary heat exchange fluid through the first heat exchanger **206** may also be redirected around the first heat exchanger **206**.

(103) In this way, heat exchanger assembly **200** may be reconfigured in response to a sensed or detected failure (e.g., a leak or broken pipe). As such, heat exchanger assembly **200** (and thermal

management system **100** of FIG. 2) may provide health monitoring to monitor for a failed air pipe due to a sensed step change or unexpected performance value of a performance parameter/characteristic (e.g., a pressure, a temperature, or other performance characteristic of a working fluid or of a component connected to heat exchanger assembly **200** (or connected to thermal management system **100**)).

(104) It will be appreciated that although in the exemplary embodiment depicted in FIGS. 2 through 4 generally discuss use of a thermal transport bus having a plurality of heat exchangers adding heat to an intermediary heat exchange fluid and extracting heat from the intermediary heat exchange fluid to manage thermal loads within an engine, other configurations are contemplated as well. For example, in other exemplary embodiments, the thermal transport bus may be a closed loop system having a relatively small sensing line flowing an intermediary heat exchange fluid. With such a configuration, the sensing line may still accept heat from one or more locations (e.g., via heat exchangers) and may further still reject heat to one or more other locations (e.g., via heat exchangers), however such an operation may have the primary goal of sensing a fault condition instead of managing thermal loads within the engine.

(105) FIG. 5 is a schematic view of a thermal management system **100'** in accordance with an exemplary aspect of the present disclosure. The embodiment provided in FIG. 5 may be configured in substantially the same manner as described in regard to FIG. 2, with the addition of a first variable volume device **168A** and a second variable volume device **168B**.

(106) A thermal management system for turbofan engine **12** (FIG. 1) includes thermal transport bus **102**, a control system engaged with thermal transport bus **102**, and a variable volume device (or more specifically for the embodiment depicted, first variable volume device **168A** and second variable volume device **168B**).

(107) Thermal transport bus **102** includes first flow loop **104**, heat exchanger **106c**, second flow loop **118**, and an intermediary heat exchange fluid flowing through thermal transport bus **102**. First flow loop **104** is fluidly isolated from second flow loop **118**. In this exemplary embodiment, heat exchanger **106c** is disposed along and in fluid communication with first flow loop **104**. It will be appreciated, however, that in other exemplary embodiments, heat exchangers **106d-e** may also be disposed along and in fluid communication with first flow loop **104**. Thermal transport bus **102** also includes heat exchanger **106c** disposed along and in fluid communication with second flow loop **118**.

(108) In certain exemplary embodiments, the pressure in first flow loop **104**, in second flow loop **118**, or in both first flow loop **104** and second flow loop **118** can be pressurized by a fuelhydraulic actuator and/or by a pressure of a general fuel system, either of which may be fluidly connected to first flow loop **104**, to second flow loop **118**, or to both first flow loop **104** and second flow loop **118**.

(109) The control system is engaged with thermal transport bus **102** and includes sensors **112** and controller **164**. Sensors **112** are operably coupled to thermal transport bus **102** and are configured to measure a temperature, a pressure, a flowrate, or a combination thereof of the intermediary heat exchange fluid. Controller **164** is connected (e.g., via a wired and/or wireless connection) to and is configured to receive electrical signals from sensors **112**.

(110) First variable volume device **168A** and second variable volume device **168B** are devices configured to increase or decrease a volume therein. In certain exemplary embodiments, first variable volume device **168A** and/or second variable volume device **168B** may include an accumulator, an actuator, a bellows, a hydraulic cylinder, a spring-loaded element, a thermal actuated element, or any combination thereof. First variable volume device **168A** and second variable volume device **168B** are in fluid communication with thermal transport bus **102**.

(111) More specifically, in at least certain exemplary aspects, first variable volume device **168A** may include a first housing **171A** defining a first internal chamber **173A**, a first piston **175A** disposed within first housing **171A** and further defining first internal chamber **173A**, and a first

actuator **177A** disposed to move first piston **175A** relative to first housing **171A**. First internal chamber **173A** is in fluid communication with thermal transport bus **102**. In an exemplary embodiment, first variable volume device **168A** may be configured to adjust the flow volume of thermal transport bus **102** by 2% to 20% of an initial flow volume of thermal transport bus **102**. In another exemplary embodiment, first variable volume device **168A** may be configured to adjust the flow volume of first flow loop **104** by 2% to 20% of an initial flow volume of first flow loop **104**.

(112) Likewise, in at least certain exemplary aspects, second variable volume device **168B** may include a second housing **171B** defining a second internal chamber **173B**, a second piston **175B** disposed within second housing **171B** and further defining the second internal chamber **173B**, and a second actuator **177B** disposed to move second piston **175B** relative to second housing **171B**. Second internal chamber **173B** is in fluid communication with thermal transport bus **102**. In an exemplary embodiment, second variable volume device **168B** may be configured to adjust the flow volume of thermal transport bus **102** by 2% to 20% of an initial flow volume of thermal transport bus **102** (e.g., a minimum flow volume of the thermal transport bus **102**). In another exemplary embodiment, second variable volume device **168B** may be configured to adjust the flow volume of first flow loop **104** by 2% to 20% of an initial flow volume of first flow loop **104** (e.g., a minimum flow volume of the first flow loop **104**).

(113) First variable volume device **168A** and second variable volume device **168B** are configured to regulate a pressure of the intermediary heat exchange fluid disposed in first flow loop **104** of thermal transport bus **102**.

(114) First variable volume device **168A** and second variable volume device **168B** are disposed to adjust a pressure of the intermediary heat exchange fluid within thermal transport bus **102**. In certain exemplary embodiments, first variable volume device **168A** and second variable volume device **168B** may be configured to adjust the volume of their respective internal chambers (e.g., first internal chamber **173A** and second internal chamber **173B**) in response to a change in pressure of the intermediary heat exchange fluid. It will be appreciated, however, that in other exemplary embodiments, first variable volume device **168A** and second variable volume device **168B** may be configured to adjust the volume of the internal chamber in response to a signal from controller **164**.

(115) In certain exemplary embodiments, first variable volume device **168A** and second variable volume device **168B** is electrically connected to controller **164**. Additionally, or alternatively, first variable volume device **168A** and second variable volume device **168B** may be in wireless communication with controller **164**.

(116) Furthermore, referring still to FIG. 5, it will be appreciated that in certain exemplary embodiments, such as the exemplary embodiment depicted, the thermal management system **100'** may additionally or alternatively include variable volume devices at other locations. For example, in the embodiment depicted, the thermal management system **100'** further includes a third variable volume device **168C** in fluid communication with the second flow loop **118** of the thermal transport bus **102**. In the embodiment depicted, the third variable volume device **168C** is located fluidly between first heat exchanger **106c** and second heat exchanger **120b**. The third variable volume device **168C** may be configured in a similar manner as first variable volume device **168A**.

(117) However, in alternative embodiments, the thermal management system **100'** additionally or alternatively includes additional or alternative variable volume device(s) at a location **169A** or at a location **169B**.

(118) A thermal management system in accordance with an exemplary embodiment present in this disclosure provides a way of controlling, actively or passively, the volume on the intermediary working fluid inside of thermal transport bus **102** in response to a pressure (or change therein) associated with either first flow loop **104** or second flow loop **118** depending upon an operating condition of turbofan engine **12** (see e.g., FIG. 1).

(119) It will be appreciated that the exemplary thermal management system **100'** depicted in FIG. 5 and described above is provided by way of example only. In other exemplary embodiments, the

thermal management system **100'** may include any other suitable number and/or configuration of variable volume device(s) **168**. For example, in certain exemplary embodiments, the thermal management system **100'** may include a single variable volume device **168** fluidly coupled to the first flow loop **104**. Additionally, or alternatively, the thermal management system **100'** may include more than two variable volume devices **168** fluidly coupled to the first flow loop **104**, may include one or more variable volume device(s) **168** fluidly coupled to the second flow loop **118**, etc. Moreover, in certain exemplary embodiments, the exemplary thermal management system **100'** may not include variable volume device(s) **168**.

(120) FIG. **6** is a flowchart of a method **300** of regulating pressure in a thermal transport bus (e.g., thermal transport bus **102** of either thermal management system **100** or thermal management system **100'**) of turbofan engine **12** (FIG. **1**) in accordance with an exemplary aspect of the present disclosure. Method **300** of regulating pressure in a thermal transport bus of a gas turbine engine includes steps **302** through **324**.

(121) Method **300** includes at **302** operating turbofan engine **12** (FIG. **1**) with thermal transport bus **102** having an intermediary heat exchange fluid flowing therethrough, thermal transport bus **102** including one or more heat source heat exchangers and one or more heat sink heat exchangers (e.g., heat exchangers **106a-e** and heat exchangers **120a-b**, respectively) in thermal communication with the intermediary heat exchanger fluid.

(122) Method **300** includes at **304** monitoring a performance characteristic of the intermediary heat exchange fluid in thermal transport bus **102** with sensors **112**. In certain exemplary embodiments, the performance characteristic may include a temperature, a pressure, a flowrate, or a combination thereof of the intermediary heat exchange fluid.

(123) Method **300** includes at **306** receiving a data set representative of the monitored performance characteristic.

(124) Method **300** includes at **308** determining, from the data set, whether the monitored performance characteristic of the intermediary heat exchange fluid is outside of a predetermined range.

(125) Method **300** includes at **310** indicating a fault condition with controller **164** if the monitored performance characteristic of the intermediary heat exchange fluid is outside of the predetermined range. For example, determining, from the data set, whether the monitored performance characteristic of the intermediary heat exchange fluid is outside of the predetermined range at **308** may include determining, from the data set, that the monitored performance characteristic of the intermediary heat exchange fluid is outside of the predetermined range. In such a case, indicating the fault condition with controller at **310** may include indicating the fault condition in response to determining the monitored performance characteristic of the intermediary heat exchange fluid is outside of the predetermined range.

(126) Method **300** includes at **312** adjusting a flow volume of thermal transport bus **102** using first variable volume device **168A**, second variable volume device **168B** or both that are in fluid communication with thermal transport bus **102**. In certain exemplary embodiments, adjusting the flow volume of thermal transport bus **102** is in response to the indicated fault condition.

(127) Method **300** may also include at **312** step **314** of adjusting the flow volume of thermal transport bus **102** using first variable volume device **168A**, second variable volume device **168B** or both that are in fluid communication with thermal transport bus **102** in response to a pressure change associated with thermal transport bus **102**.

(128) Method **300** may also include at **312** step **316** of redirecting a flow of the intermediary heat exchange fluid to a redundant flow loop (e.g., from first flow loop **104** to second flow loop **118**) of thermal transport bus **102**. In an exemplary embodiment, adjusting the flow volume of thermal transport bus **102** is in response to redirecting the intermediary heat exchange fluid to the redundant flow loop (e.g., from first flow loop **104** to second flow loop **118**).

(129) Method **300** may also include at **312** step **318** of redirecting a flow of the intermediary heat

exchange fluid from a redundant flow loop (e.g., to first flow loop **104** from second flow loop **118**) of thermal transport bus **102**. In another exemplary embodiment, adjusting the flow volume of thermal transport bus **102** is in response to redirecting the intermediary heat exchange fluid from the redundant flow loop (e.g., to first flow loop **104** from second flow loop **118**, or both).

(130) Method **300** may also include at **312** step **320** of varying a size of first internal chamber **173A** of first variable volume device **168A**, a size second internal chamber **173B** of second variable volume device **168B**, or a size of both first and second internal chambers **173A** and **173B**.

(131) Method **300** may also include at **312** step **322** of determining data indicative of an operating pressure of the intermediary heat exchange fluid. In an exemplary embodiment, adjusting the flow volume of thermal transport bus **102** may include adjusting the flow volume of thermal transport bus **102** in response to determining data indicative of the operating pressure of the intermediary heat exchange fluid.

(132) Method **300** may also include at **312** step **324** of automatically adjusting the flow volume of thermal transport bus **102** in response to the pressure change within thermal transport bus **102**.

(133) It will be appreciated that the exemplary aspect of the method **300** depicted in FIG. **6** and discussed above is provided by way of example only. In other exemplary aspects, the method **300** may not include each of the steps described herein and depicted in FIG. **6**. Additionally, or alternatively, in other exemplary aspects, the method **300** may include additional steps not described herein or depicted in FIG. **6**.

(134) A thermal management system in accordance with an exemplary embodiment present in this disclosure may allow for a gas turbine engine and aircraft including such a gas turbine engine to operate more efficiently. More particularly, inclusion of a thermal management system in accordance with an exemplary embodiment of the present disclosure may allow for a quicker response to leak or failure conditions of the gas turbine engine or the aircraft if they occur. Additionally, the heat exchange fluid used by a thermal management system in accordance with an exemplary embodiment of the present disclosure may be redirected by actively disengaging a heat exchanger in response to a detected leak or failure in a component fluidly connected to one of the heat exchangers.

(135) This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

(136) Further aspects are provided by the subject matter of the following clauses:

(137) A method of detecting an airflow fault condition in a gas turbine engine, the method comprising: operating the gas turbine engine with a thermal transport bus having an intermediary heat exchange fluid flowing therethrough; determining a performance characteristic of the intermediary heat exchange fluid in the thermal transport bus is outside of a predetermined range, wherein the performance characteristic comprises a temperature, a pressure, a flowrate, or a combination thereof; and indicating an airflow fault condition in response to determining the performance characteristic is outside of the predetermined range.

(138) The method of one or more of these clauses, further comprising: modifying a configuration of airflow in the gas turbine engine in response to the indicated airflow fault condition.

(139) The method of one or more of these clauses, wherein modifying the configuration of airflow in the gas turbine engine comprises: selectively deactivating a first heat exchanger in fluid communication with the thermal transport bus; and selectively bypassing the first heat exchanger by sending a flow of air from the first heat exchanger to a second heat exchanger in fluid

communication with the thermal transport bus.

(140) The method of one or more of these clauses, further comprising: monitoring the performance characteristic of the intermediary heat exchange fluid, wherein monitoring the performance characteristic of the intermediary heat exchange fluid comprises sensing data indicative of the performance characteristic of the intermediary heat exchange fluid with a sensor, wherein sensing data indicative of the performance characteristic of the intermediary heat exchange fluid comprises sensing a piecewise change in the data indicative of the performance characteristic of the intermediary heat exchange fluid.

(141) The method of one or more of these clauses, further comprising: monitoring the performance characteristic of the intermediary heat exchange fluid, wherein monitoring the performance characteristic of the intermediary heat exchange fluid comprises measuring the performance characteristic with a temperature sensor, a pressure sensor, or both.

(142) The method of one or more of these clauses, wherein the performance characteristic is a differential temperature, a differential pressure, a differential flowrate, or a combination thereof.

(143) The method of one or more of these clauses, further comprising:

(144) modifying a configuration of the thermal transport bus in response to the indicated airflow fault condition by varying a position of a valve in fluid communication with the thermal transport bus.

(145) The method of one or more of these clauses, wherein modifying the configuration of the thermal transport bus comprises modifying a configuration of a secondary circuit of the gas turbine engine by varying the position of the valve in fluid communication with the secondary circuit.

(146) The method of one or more of these clauses, further comprising: monitoring the performance characteristic of the intermediary heat exchange fluid, wherein monitoring the performance characteristic of the intermediary heat exchange fluid comprises monitoring for an instantaneous change in the performance characteristic.

(147) The method of one or more of these clauses, wherein the instantaneous change in the performance characteristic is defined by a step function of the performance characteristic.

(148) The method of one or more of these clauses, further comprising: creating a data set representative of the performance characteristic of the intermediary heat exchange fluid, wherein determining the performance characteristic is outside of the predetermined range comprises identifying a portion of the data defining a step function.

(149) The method of one or more of these clauses, further comprising: monitoring the performance characteristic of the intermediary heat exchange fluid, wherein monitoring the performance characteristic of the intermediary heat exchange fluid comprises monitoring for a gradual change in the performance characteristic.

(150) The method of one or more of these clauses, wherein the airflow fault condition is indicative of a broken pipe of the gas turbine engine.

(151) The method of one or more of these clauses, wherein operating the gas turbine engine with the thermal transport bus comprises: providing the intermediary heat exchange fluid to a heat exchanger; and providing a fluid flow through a duct of the gas turbine engine to the heat exchanger to exchange heat with the intermediary heat exchange fluid; wherein the airflow fault condition is a fault associated with the duct of the gas turbine engine, and wherein modifying the configuration of airflow in the gas turbine engine comprises modifying the airflow through the duct of the gas turbine engine.

(152) The method of one or more of these clauses, wherein the thermal transport bus is configured to connect with a heat source heat exchanger, wherein the heat source heat exchanger comprises a cooled cooling air heat exchanger, a waste heat recovery heat exchanger, an air cooled oil cooler heat exchanger, or any combination thereof.

(153) The method of one or more of these clauses, wherein the thermal transport bus is configured to connect with a heat sink heat exchanger, wherein the heat sink heat exchanger comprises a fuel

cooled bus cooler heat exchanger, an air cooled bus cooler heat exchanger, or any combination thereof.

(154) A gas turbine engine comprising: a turbomachine having compressor section, a combustion section, and a turbine section arranged in serial flow order; a thermal transport bus comprising one or more heat source heat exchangers, one or more heat sink heat exchangers, a bus fluidly connected to the one or more heat source heat exchangers and to the one or more heat sink heat exchangers; and a controller operably coupled to the turbomachine, the controller comprising one or more processors and memory, the memory storing instructions that when executed by the controller cause the gas turbine engine to perform the following operations: operate the gas turbine engine with the thermal transport bus having an intermediary heat exchange fluid flowing therethrough; determine a performance characteristic of the intermediary heat exchange fluid in the thermal transport bus is outside of a predetermined range, wherein the performance characteristic comprises a temperature, a pressure, or both; and indicate an airflow fault condition in response to determining the performance characteristic is outside of the predetermined range.

(155) The gas turbine engine of one or more of these clauses, wherein the one or more heat source heat exchangers comprises a cooled cooling air heat exchanger, a waste heat recovery heat exchanger, an air cooled oil cooler heat exchanger, or any combination thereof.

(156) The gas turbine engine of one or more of these clauses, wherein the one or more heat sink heat exchangers comprises a fuel cooled bus cooler heat exchanger, an air cooled bus cooler heat exchanger, or any combination thereof.

(157) The gas turbine engine of one or more of these clauses, further comprising: a sensor connected to the bus and configured to communicate with the controller, wherein the sensor is further configured to monitor the performance characteristic of the heat exchange fluid; and one or more valves connected to the bus and configured to receive signals from the controller, wherein the one or more valves are further configured to modifying the configuration of airflow in the gas turbine engine.

Claims

1. A method of detecting an airflow fault condition in a gas turbine engine, the method comprising: operating the gas turbine engine with a thermal transport bus, the thermal bus including a first flow loop including a first compressor, the first flow loop configured to circulate a first intermediary heat exchange fluid flowing therethrough, wherein the first flow loop is a closed loop conduit, and a second flow loop including a second compressor, the second flow loop configured to circulate a second intermediary heat exchange fluid flowing therethrough, wherein the second flow loop is a closed loop conduit, wherein the first flow loop is fluidly isolated from and redundant to the second flow loop, wherein the first flow loop and the second flow loop are thermally coupled to a common heat source, wherein the first compressor and the second compressor are driven by a common drive system; determining a performance characteristic of the first intermediary heat exchange fluid in the thermal transport bus is outside of a predetermined range, wherein the performance characteristic comprises a temperature, a pressure, a flowrate, or a combination thereof; and indicating an airflow fault condition in response to determining the performance characteristic is outside of the predetermined range.

2. The method of claim 1, further comprising: modifying a configuration of airflow in the gas turbine engine in response to the indicated airflow fault condition.

3. The method of claim 2, wherein modifying the configuration of airflow in the gas turbine engine comprises: selectively deactivating a first heat exchanger in fluid communication with the thermal transport bus; and selectively bypassing the first heat exchanger by sending a flow of air from the first heat exchanger to a second heat exchanger in fluid communication with the thermal transport bus.

4. The method of claim 1, further comprising: monitoring the performance characteristic of the first intermediary heat exchange fluid, wherein monitoring the performance characteristic of the first intermediary heat exchange fluid comprises sensing data indicative of the performance characteristic of the first intermediary heat exchange fluid with a first sensor, wherein sensing data indicative of the performance characteristic of the first intermediary heat exchange fluid comprises sensing a piecewise change in the data indicative of the performance characteristic of the first intermediary heat exchange fluid.
 5. The method of claim 1, further comprising: monitoring the performance characteristic of the first intermediary heat exchange fluid, wherein monitoring the performance characteristic of the first intermediary heat exchange fluid comprises measuring the performance characteristic with a temperature sensor, a pressure sensor, or both.
 6. The method of claim 1, wherein the performance characteristic is a differential temperature, a differential pressure, a differential flowrate, or a combination thereof.
 7. The method of claim 1, further comprising: modifying a configuration of the thermal transport bus in response to the indicated airflow fault condition by varying a position of a valve in fluid communication with the thermal transport bus.
 8. The method of claim 7, wherein modifying the configuration of the thermal transport bus comprises modifying a configuration of a secondary circuit of the gas turbine engine by varying the position of the valve in fluid communication with the secondary circuit.
 9. The method of claim 1, further comprising: monitoring the performance characteristic of the first intermediary heat exchange fluid, wherein monitoring the performance characteristic of the first intermediary heat exchange fluid comprises monitoring for an instantaneous change in the performance characteristic.
 10. The method of claim 9, wherein the instantaneous change in the performance characteristic is defined by a step function of the performance characteristic.
 11. The method of claim 1, further comprising: creating a data set representative of the performance characteristic of the first intermediary heat exchange fluid, wherein determining the performance characteristic is outside of the predetermined range comprises identifying a portion of the data defining a step function.
 12. The method of claim 1, further comprising: monitoring the performance characteristic of the first intermediary heat exchange fluid, wherein monitoring the performance characteristic of the first intermediary heat exchange fluid comprises monitoring for a gradual change in the performance characteristic.
 13. The method of claim 1, wherein the airflow fault condition is indicative of a broken pipe of the gas turbine engine.
 14. The method of claim 1, wherein operating the gas turbine engine with the thermal transport bus comprises: providing the first intermediary heat exchange fluid to a heat exchanger; and providing a fluid flow through a duct of the gas turbine engine to the heat exchanger to exchange heat with the first intermediary heat exchange fluid; wherein the airflow fault condition is a fault associated with the duct of the gas turbine engine, and wherein modifying the configuration of airflow in the gas turbine engine comprises modifying the airflow through the duct of the gas turbine engine.
 15. The method of claim 1, the common heat source comprises a cooled cooling air heat exchanger, a waste heat recovery heat exchanger, an air cooled oil cooler heat exchanger, or any combination thereof.
 16. The method of claim 1, wherein the first flow loop and the second flow loop of the thermal transport bus are configured to connect with a heat sink heat exchanger, wherein the heat sink heat exchanger comprises a fuel cooled bus cooler heat exchanger, an air cooled bus cooler heat exchanger, or any combination thereof.
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